Circuit Theory Foundations — Problem Set

Sections 1–13, cross-referenced to ECE 105 Lectures 5–7

Conceptual, analytical, and Python problems with full solutions, grouped by topic

ECE 105 — Introduction to Electrical Engineering

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Problems are grouped by topic. Full solutions, in matching order, begin on page 4. Unless stated otherwise, treat all resistors as ideal and all sources as ideal.

Problems

A. Charge, Current, Voltage, and Power

Problem 1 (conceptual). Explain the difference between conventional current and electron flow. If a formula gives a current of $+2\,\mathrm{A}$ to the right, which way are the electrons actually drifting?

Problem 2 (analytical). A device carries a steady $0.40\,\mathrm{A}$ for 5 minutes. (a) How much charge passes? (b) How many electrons is that?

Problem 3 (analytical). A $9\,\mathrm{V}$ battery delivers $150\,\mathrm{mA}$ to a circuit. (a) What power does it supply? (b) How much energy (in joules) does it deliver in 10 minutes?

Problem 4 (conceptual). Voltage is defined "between two points," yet we routinely say "the voltage at node A is $3\,\mathrm{V}$." What unstated convention makes that statement meaningful?

B. Resistance and Ohm's Law

Problem 5 (conceptual). State the difference between the *definition* $R = V/I$ and *Ohm's law* $V = IR$. For which kind of element does only the first one apply?

Problem 6 (analytical). A resistor drops $3.3\,\mathrm{V}$ when $15\,\mathrm{mA}$ flows through it. (a) Find R. (b) Find the power it dissipates. (c) Would a $\frac{1}{8}\,\mathrm{W}$ resistor survive?

Problem 7 (analytical). Two wires are the same material and length, but wire B has twice the cross-sectional area of wire A. What is the ratio R_B/R_A? If wire A is instead stretched to twice its length (area unchanged), what is its new resistance relative to the original?

Problem 8 (Python). Write a short script that plots I versus V for a $470\,\Omega$ resistor over $V \in [0, 5]\,\mathrm{V}$, and on the same axes plots the (nonohmic) relation $I = I_s(e^{V/V_T} - 1)$ for a diode with $I_s = 10^{-12}\,\mathrm{A}$, $V_T = 25.85\,\mathrm{mV}$. Comment on why only one of the two is a straight line.

C. Series, Parallel, and the Dividers

Problem 9 (analytical). Find the equivalent resistance of $1\,\mathrm{k\Omega}$, $2\,\mathrm{k\Omega}$, and $2\,\mathrm{k\Omega}$ all in parallel.

Problem 10 (analytical). A voltage divider has $V_{\text{in}} = 12\,\mathrm{V}$, $R_1 = 8.2\,\mathrm{k\Omega}$ (top) and $R_2 = 3.3\,\mathrm{k\Omega}$ (bottom). (a) Find V_{out} (across R_2). (b) Now connect a $3.3\,\mathrm{k\Omega}$ load across the output. Recompute V_{out} and comment on the change.

Problem 11 (analytical). A $10\,\mathrm{mA}$ source feeds two parallel branches, $R_1 = 1\,\mathrm{k\Omega}$ and $R_2 = 4\,\mathrm{k\Omega}$. Use the current-divider rule to find the current in each branch. Which branch carries more, and why?

Problem 12 (conceptual). Explain in one or two sentences why the parallel combination of two resistors is always smaller than either one.

D. Sources, Measurement, and Faults

Problem 13 (conceptual). A real $1.5\,\mathrm{V}$ battery has internal resistance $r_s = 0.5\,\Omega$. Sketch (in words) why its terminal voltage drops as the load draws more current, and compute the terminal voltage into a $2.5\,\Omega$ load.

Problem 14 (conceptual). Why is an ammeter placed in series and a voltmeter in parallel? What input resistance would each ideally have, and what goes wrong if a real meter departs from the ideal?

Problem 15 (conceptual). Distinguish a short circuit from an open circuit in terms of resistance and resulting current, and explain how a fuse uses one to prevent damage from the other.

E. Kirchhoff's Laws and Nodal Analysis

Problem 16 (analytical). A $10\,\mathrm{V}$ source drives three resistors in series: $R_1 = 1\,\mathrm{k\Omega}$, $R_2 = 2\,\mathrm{k\Omega}$, $R_3 = 2\,\mathrm{k\Omega}$. Use KVL to find the loop current and the voltage across each resistor. Verify the drops sum to $10\,\mathrm{V}$.

Problem 17 (analytical). A $12\,\mathrm{V}$ source connects through $R_1 = 2\,\mathrm{k\Omega}$ to node A. From A, $R_2 = 4\,\mathrm{k\Omega}$ goes to ground and $R_3 = 4\,\mathrm{k\Omega}$ goes to ground. Use nodal analysis to find V_A and the source current.

Problem 18 (Python). Set up the nodal equations for a two-node resistive network of your choice as a matrix equation $\mathbf{Gv} = \mathbf{i}$ and solve it with `numpy.linalg.solve`. Print the node voltages and verify KCL numerically at each node.

F. Superposition, Thévenin, and Norton

Problem 19 (conceptual). When applying superposition, how do you "turn off" a voltage source? A current source? Why does the method fail if the circuit contains a diode?

Problem 20 (analytical). For the divider $V_{\text{in}} = 9\,\mathrm{V}$, $R_1 = 4.7\,\mathrm{k\Omega}$ (top), $R_2 = 10\,\mathrm{k\Omega}$ (bottom), output across R_2: find the Thévenin equivalent (V_{TH}, R_{TH}) seen at the output terminals, then use it to predict V_{out} when a $10\,\mathrm{k\Omega}$ load is attached.

Problem 21 (analytical). Convert the Thévenin equivalent from the previous problem into its Norton equivalent (I_{N}, R_{TH}). Confirm $V_{\text{TH}} = I_{\text{N}} R_{\text{TH}}$.

G. Capacitors and Inductors

Problem 22 (analytical)**.** A $2.2\,\mu$F capacitor charges through a $22\,\text{k}\Omega$ resistor from a step input. (a) Find the time constant. (b) Roughly how long until it is $> 99\%$ charged?

Problem 23 (conceptual)**.** State the duality between capacitors and inductors. Which quantity can "not jump" for each element, and which defining equation expresses that?

Solutions

Solution 1. Conventional current is the direction *positive* charge would move; electron flow is opposite. A $+2\,\mathrm{A}$ current to the right means electrons drift to the *left*.

Solution 2. (a) $Q = I\,t = 0.40 \times 300 = 120\,\mathrm{C}$. (b) $120/(1.602 \times 10^{-19}) \approx 7.5 \times 10^{20}$ electrons.

Solution 3. (a) $P = VI = 9 \times 0.150 = 1.35\,\mathrm{W}$. (b) $E = P\,t = 1.35 \times 600 = 810\,\mathrm{J}$.

Solution 4. A single node voltage is understood *relative to a chosen* $0\,\mathrm{V}$ *reference* (ground). "$V_A = 3\,\mathrm{V}$" means the potential difference between A and ground is $3\,\mathrm{V}$.

Solution 5. $R = V/I$ is the *definition* of resistance and holds for any element point-by-point. $V = IR$ with constant R (Ohm's law) holds only for *ohmic* elements. For a nonohmic element (diode, filament), only the definition applies.

Solution 6. (a) $R = V/I = 3.3/0.015 = 220\,\Omega$. (b) $P = VI = 3.3 \times 0.015 = 49.5\,\mathrm{mW}$. (c) A $\frac{1}{8}\,\mathrm{W} = 125\,\mathrm{mW}$ resistor is rated well above $49.5\,\mathrm{mW}$, so yes.

Solution 7. $R \propto L/A$. Doubling area halves resistance: $R_B/R_A = 0.5$. Doubling length (area fixed) doubles resistance; if the wire is stretched so volume is conserved the area would halve and R would rise $\times 4$, but with area held fixed the answer is $\times 2$.

Solution 8. The resistor line is straight because $I = V/R$ is linear (constant slope $1/R$). The diode's exponential $I(V)$ curves sharply upward, so it is nonohmic — no single resistance describes it.

```
import numpy as np, matplotlib.pyplot as plt
V = np.linspace(0, 5, 400)
I_res = V/470.0
Is, VT = 1e-12, 0.02585
I_diode = Is*(np.exp(V/VT)-1)
plt.plot(V, I_res*1e3, label='470 ohm (ohmic)')
plt.plot(V, np.clip(I_diode,0,0.05)*1e3, label='diode (nonohmic)')
plt.xlabel('V (V)'); plt.ylabel('I (mA)'); plt.legend(); plt.show()
```

Solution 9. $1/R = 1/1000 + 1/2000 + 1/2000 = 0.002\,\mathrm{S}$, so $R = 500\,\Omega$.

Solution 10. (a) $V_{\mathrm{out}} = 12{\cdot}3.3/(8.2{+}3.3) = 12{\cdot}0.287 = 3.44\,\mathrm{V}$. (b) Load $3.3\,\mathrm{k\Omega} \parallel 3.3\,\mathrm{k\Omega} = 1.65\,\mathrm{k\Omega}$; $V_{\mathrm{out}} = 12 \cdot 1.65/(8.2 + 1.65) = 12 \cdot 0.168 = 2.01\,\mathrm{V}$. The output sags because the divider is not a stiff source — loading it lowers the bottom-leg resistance.

Solution 11. $I_1 = I\,R_2/(R_1 + R_2) = 10 \cdot 4/(1 + 4) = 8\,\mathrm{mA}$; $I_2 = 10 \cdot 1/5 = 2\,\mathrm{mA}$. The *smaller* resistor (R_1) carries the *larger* current — current prefers the easier path.

Solution 12. Adding a parallel path gives the current more ways to flow, which can only increase total conductance; larger conductance means smaller resistance than either branch alone.

Solution 13. Terminal voltage $= V_S\,R_L/(R_L + r_s) = 1.5 \cdot 2.5/(2.5 + 0.5) = 1.25\,\mathrm{V}$. As R_L falls, more current flows, so more voltage is dropped across r_s, leaving less at the terminals.

Solution 14. An ammeter must carry the branch current, so it goes in series and ideally has $0\,\Omega$ (no added drop). A voltmeter must see the potential difference, so it goes in parallel and ideally has infinite resistance (no stolen current). A real meter's non-ideal resistance loads the circuit and biases the reading, worst when comparable to the circuit resistance.

Solution 15. A short is $R \to 0$, giving very large current ($I = V/R$); an open is $R \to \infty$, giving zero current. A fuse carries normal current but melts into an *open* when current gets too large (e.g. from a short), interrupting the circuit before damage spreads.

Solution 16. $I = 10/(1000 + 2000 + 2000) = 2\,\text{mA}$. Drops: $IR_1 = 2\,\text{V}$, $IR_2 = 4\,\text{V}$, $IR_3 = 4\,\text{V}$; sum $= 10\,\text{V}$. ✓

Solution 17. KCL at A: $(V_A - 12)/2\text{k} + V_A/4\text{k} + V_A/4\text{k} = 0$. $\times 4\text{k}$: $2(V_A - 12) + V_A + V_A = 0 \Rightarrow 4V_A = 24 \Rightarrow V_A = 6\,\text{V}$. Source current $= (12 - 6)/2\text{k} = 3\,\text{mA}$.

Solution 18. Example network (node voltages v_1, v_2):

```
import numpy as np
# Conductance matrix G v = i (siemens, amps)
G = np.array([[1/1000+1/2000, -1/2000],
              [-1/2000, 1/2000+1/4000]])
i = np.array([12/1000, 0.0]) # 12 V source via 1k into node 1
v = np.linalg.solve(G, i)
print("node voltages:", v)
```

Substituting v back into each row of $\mathbf{Gv} - \mathbf{i}$ should give ≈ 0, confirming KCL.

Solution 19. Turn off a voltage source by replacing it with a *short* ($0\,\text{V}$); turn off a current source by replacing it with an *open* ($0\,\text{A}$). Superposition needs linearity ($V = IR$ with constant R); a diode is nonlinear, so responses do not add and the method fails.

Solution 20. V_{TH} = open-circuit voltage $= 9 \cdot 10/(4.7 + 10) = 6.12\,\text{V}$. $R_{\text{TH}} = R_1 \parallel R_2 = (4.7 \cdot 10)/(4.7 + 10) = 3.20\,\text{k}\Omega$ (source shorted). With a $10\,\text{k}\Omega$ load: $V_{\text{out}} = 6.12 \cdot 10/(10 + 3.20) = 4.64\,\text{V}$.

Solution 21. $I_{\text{N}} = V_{\text{TH}}/R_{\text{TH}} = 6.12/3200 = 1.91\,\text{mA}$, with the same $R_{\text{TH}} = 3.20\,\text{k}\Omega$ in parallel. Check: $I_{\text{N}} R_{\text{TH}} = 1.91\,\text{mA} \times 3.20\,\text{k}\Omega = 6.12\,\text{V} = V_{\text{TH}}$. ✓

Solution 22. (a) $\tau = RC = 22\,000 \times 2.2 \times 10^{-6} = 48.4\,\text{ms}$. (b) $> 99\%$ charged after about $5\tau \approx 242\,\text{ms}$.

Solution 23. Capacitor and inductor are duals ($V \leftrightarrow I$, $C \leftrightarrow L$). A capacitor's *voltage* cannot jump (from $I = C\,dV/dt$); an inductor's *current* cannot jump (from $V = L\,dI/dt$).